

Definition: Product is expected to deliver the shortest path to the nearest bicycle parking lot to a desired location, ensuring that the user has a bicycle parking lot to park at, given the weather, time and capacity.

Product offers routes to parking lots and filters such as sheltered parking lots, group riding and sharing of ride information for safety. Report system is also offered for users to give feedback about the capacity and availability of the parking lot.

Product focuses on user's convenience, allowing them to save time and not manually look for a parking lot. It also saves time by making sure it's the parking lot they need, for example, sheltered or not sheltered.

Data Dictionary (need to wait for requirements to be out)

- Define the terms used (users, and objects)
- Name (String) (size 20) Name of User, Eg: Linyue
- Address (String) (Size 50) Addy, Eg: 126 Sembawang Road
- Phone No (Text) (size 10)
- DOB
- Locations (altitude, latitude)

Name: User

- Type: Class
- Description: An entity that represents a human actor that interacts with this system. This entity begins by looking for a bicycle parking spot within the geographic region of Singapore.
- Source: User database
- Format: Object type consisting of phone number(s), firstName, lastName, age, gender as properties.
- Relationships: Contains saved locations (coordinates) and/or addresses that the User may use during their interactions within the system.
- Ownership: Maintained by the User. Relevant properties may only be updated by the User that owns this data.
- Access Control: Only the original User may access and/or modify the object data, potentially through SMS verification.

Name: Cyclist

- Type: Object
- Description: An instance of the User class that represents a human actor with the role of a 'cyclist', defined as someone who actively cycles or performs cycling activities within the geographic region.
- Source: User class

- Format: User class with further properties including (but not limited to) the endurance and range of the actor.
- Relationships: Behaves as an entity of User so therefore, contains relationships in the same way.
- Ownership: Similar to the User class.
- Access Control: Similar to the User class.

Name: Maps/Geographic Region, or Coordinates

- Type: API reference, Object/String
- Description: Live referencing of mapping systems using relevant web APIs.
- Source: Owners of the APIs.
- Format: JSON/XML return types with usable properties/notations.
- Relationships: Interacts with the live dataset to customize markers/information on mapping services.
- Ownership: System-referenced through a unique API key, potentially.
- Access: Fetch API with API key verification to access the services.

Name: Land Transport Authority (LTA)

- Type: Entity
- Description: The governing authority of Singapore's land transport systems.
- Source: LTA's DataMall
- Format: See above.

Functional Requirements:

Locate Nearest Bicycle Parking Lot:

- Users should be able to initiate a search for the nearest bicycle parking lot to the user by providing their current coordinates.
- The system will utilize these coordinates to identify the closest parking facility to them based

Find Bicycle Parking Lot Near Specified Location:

- Users can input the coordinates of a desired location they intend to visit.
- Upon submission, the system will determine the nearest bicycle parking lot to the specified destination.

Display Default Top 3 Results:

- Upon search initiation, the system will automatically present the top 3 nearest bicycle parking lots.
- The system map display will display the location of these top 3 nearest bicycle lots.

Option to See More Results:

- Users can opt to view additional bicycle parking options beyond the initial 3 by selecting a designated "See More" button.

Detailed Information Display:

- Upon selecting a specific parking lot, users will be presented with comprehensive details such as slot availability and shelter availability of the specific parking lot.

Filtering Options:

- Users can filter bicycle parking lots based on attributes such as availability, and shelter availability.
- The system will then display only the parking options that meet the specified criteria to the user.

Select Destination:

- After reviewing available options, users can select a preferred bicycle parking lot as their destination.

Map Display:

- A dynamic map display will showcase the user's current location, the destination bicycle parking lot, and other bicycle parking options.

Integration with Google Maps:

- The system will fetch live data from Google map and then display the map accordingly on the map display.
- The system will use Google Maps GPS information to design the navigation directions for the user.

Navigation Directions:

- Upon selecting a bicycle parking lot as the destination, users will receive navigation directions from their current location to destination

Save Common Inputs:

- Users can save frequently visited locations such as home and workplace for quick access in future searches.
- The system will have a database to save these user inputs for each user.

Parking Lot Occupancy Reporting:

- Users have the option to report the occupancy status of a bicycle parking lot, indicating whether it is fully occupied or not.

Real-Time Updates:

- Users can contribute real-time updates on parking lot occupancy, helping to maintain accurate and up-to-date information within the system.

Integration with LTA API:

- The system will integrate with the Land Transport Authority (LTA) API to fetch live data regarding bicycle parking lots.

Non-functional Requirements:

Performance:

- Response time for fetching and displaying parking lot information should be within 3 seconds.

Usability:

- The user interface should be intuitive and easy to navigate for more seamless user experience.
- Information presented to users should be clear, concise, and easily understandable in their language of choice.
- The system should support accessibility features to accommodate users with reading disabilities such as text to speech.

Reliability:

- The system should operate reliably under normal usage conditions, with minimal downtime or disruptions at a maximum of 0.5% downtime per day.
- Data presented to users should be accurate and up-to-date based on the latest update on LTA API and google maps, reflecting real-time information about parking lot availability.

Security:

- User data, including location information and saved inputs, should be securely stored and protected against unauthorized access or manipulation.
- Communication between the user's device and the system should be encrypted to ensure data privacy.

Scalability:

- The system should be able to handle a growing user base of at least Singapore's current population and increasing data volume without significant degradation in performance.

Maintainability:

- The system should be modular and well-organized, facilitating ease of maintenance and future enhancements.
- Code should be well-documented and adhere to coding standards to support ongoing development efforts.

Compatibility:

- The system should be compatible with a wide range of mobile devices and web browsers to ensure broad accessibility.
- Integration with external APIs, such as Google Maps and the LTA API, should follow industry standards and protocols.

Availability:

- The system should have high availability, with uptime of at least 99.5% to ensure continuous access for users.

Interoperability:

- The system should be interoperable with other transportation and navigation services to provide seamless user experiences.
- Integration with external APIs should be standardized to enable interoperability with third-party applications and services.

Performance Under Load:

- The system should be able to handle at least 10,000 concurrent user requests and heavy loads during peak usage periods without degradation in performance.
- Load testing should be conducted periodically to identify and address potential performance bottlenecks.

Data Privacy and Compliance:

- The system should comply with relevant data privacy regulations, such as PDPA to protect user privacy and rights.
- User consent should be obtained for collecting and processing personal data, including location information.

Use cases (need to wait for requirements to be out)

- Perform a search query.
- Split into single options (user filter for each criteria, extending from search)
- View a list of available spots.

UI:

Main page is a login page with phone number verification and pw

-> lead u to a map like gmaps with a search bar at the top and a filter bar below with options(radius, shelter, total lots, type of lot)

When user uses search bar, there's an option displayed below the bar to show saved location

-> when he clicks on it, it will show a list of saved locations and when he selects one, the location will be inputted into the search bar

-> if not he can just key into the search bar

Filter bar will display dropdown selections for radius, shelter, type of lots, total lots(can be displayed in range, eg. 1-10, 11-20, 21-30)

Once search is done, map will display location of all applicable nearest lots on map

-> directions button bottom left, and a display current location button right

-> on GPS prompt if not on

-> if on show current location represented by dot

Left panel (once clicked) displays share location, share group ride destination



Use Case 1: Sign-up

Description: The user signs up for a new account

Actors: User, Application System

Pre-conditions: The User has a mobile number and does not have an existing account

Post-conditions: The User creates an account successfully

The flow of events:

1. The application will prompt the User for his/her mobile number.
2. The User will input his/her 10-digit mobile number, including country code (+65).
3. The application validates the mobile number.
4. Upon successful validation, the application will prompt the User for a 'password' and 'password confirmation'.
5. The User will enter the exact password for the 'password' prompt and 'password confirmation' prompt.
6. Once 'password' input and 'password confirmation' inputs are verified, the application will store the User's mobile number and password in the system.
7. A new account will be created for the User.

Alternative flows:

AF-S2: If the mobile number is invalid

1. The application will display "Invalid mobile number. Please try again!" for 1 second.
2. The application will return to step 1.

AF-S5: If the passwords do not match

1. The application will display "Passwords do not match. Please try again!" for 1 second.
2. The application will return to step 4.

Use Case 2: Login

Description: User logs in with an existing account

Actors: User, Application System

Pre-conditions: The User has an existing account

Post-conditions: The User logs into the account successfully

The flow of events:

1. The application will prompt the User for his/her mobile number.
2. The User will input his/her mobile number.
3. The application will prompt the User for his/her password.
4. The User will input his/her password.
5. The application uses the included use case "Verification" to verify the User login details.
6. Upon successful verification, the User is logged into his account.

Alternative flows:

AF-S5: If the verification is unsuccessful

1. The application will display "Incorrect mobile number or password. Please try again!" for 1 second.
2. The application will return to step 1.
3. After 3 attempts, the application will prompt the User to do a password reset.
4. The application will use the extended use case "Reset Password" if the User chooses to reset his/her password.

Includes Use case "Verification"

Extends Use case "Reset Password"

Use Case 3: Verification

Description: The application verifies the login details of the User

Actors: User, Application System

Pre-conditions: The User has an existing account, with no changes to his/her mobile number

Post-conditions: The User login details are verified successfully

The flow of events:

1. The Application System retrieves the mobile number from the User input and searches its database for a valid mobile number.
2. The Application System retrieves the password from the User input and verifies in its database if it matches the password corresponding to the mobile number.
3. If the mobile number is valid, and the password corresponds, the Application System returns Boolean 'true', and the User is successfully verified.

Alternative flows:

AF-S3: If the Application System database does not contain the mobile number or that the password entered by the User does not match the password corresponding to the mobile number entered by the User

1. The Application System will return Boolean 'false', and the User is not verified.

Use Case 4: Reset Password

Description: User resets password to an existing account

Actors: User, Application System

Pre-conditions: The User has an existing account, with no changes to his/her mobile number

Post-conditions: The User password is reset successfully

The flow of events:

1. The application will prompt the User for his mobile number.
2. The User will input his/her mobile number.
3. Upon mobile number verification, the application will prompt the User for 'new password' and 'password confirmation'.
4. The User will input the exact password for the 'new password' prompt and the 'password confirmation' prompt.
5. Once the 'new password' input and 'password confirmation' inputs are verified, the application will store the updated password in the system.
6. The password is reset for the User.

Alternative flows:

AF-S2: If the mobile number is invalid

3. The application will display "Invalid mobile number. Please try again!" for 1 second.
4. The application will return to step 1.

AF-S4: If the passwords do not match

3. The application will display "Passwords do not match. Please try again!" for 1 second.
4. The application will return to step 3.

Use Case 3: Help

Description: User enquires for help regarding privacy concerns or FAQs.

Actors: User, Application System

Pre-conditions: The User is logged into his/her account

Post-conditions: The User is displayed options to enquire regarding privacy concerns or FAQs

The flow of events:

1. The application will use the included use case “Login” to verify the User.
2. The User clicks on the ‘Help’ option in the application.
3. The application displays the option to enquire about privacy concerns or FAQs with the extended use cases “Enquire Privacy Concerns” and “FAQ”.

Extends Use case “Enquire Privacy Concerns” and “FAQ”

Use Case 4: Enquire privacy concerns

Description: User enquires about privacy concerns

Actors: User, Application System

Pre-conditions: The User is logged into his/her account

Post-conditions: The User is displayed with privacy protocols of the application

The flow of events:

1. The application will use the included use case “Login” to verify the User.
2. The User clicks on the option ‘Enquire privacy concerns’ after clicking on the ‘Help’ option in the application.
3. The application will display the privacy protocols of the application to ensure data security.

Use Case 5: FAQ

Description: User enquires for FAQs.

Actors: User, Application System

Pre-conditions: The User is logged into his/her account

Post-conditions: The User is displayed with FAQs

The flow of events:

1. The application will use the included use case “Login” to verify the User.
2. The User clicks on the option ‘FAQs’ after clicking on the ‘Help’ option in the application.
3. The application will display the list of FAQs and their respective replies.

Use Case 6: Find Bicycle Parking Lot

Description: The user is displayed 5 bicycle parking lots nearest to his/her current location or specified destination

Actors: User, Application System, Parking Lot Database, GPS, GoogleMaps

Pre-conditions: The User is logged into his/her account, with GPS enabled

Post-conditions: Locations of bicycle parking lots nearest to his/her current location or specified destination are displayed on GoogleMaps

The flow of events:

1. The application will use the included use case "Login" to verify the User.
2. The User can filter bicycle parking lot coordinates based on shelter.
3. The application will retrieve the coordinates of all filtered bicycle parking lots from the Parking Lot Database using the included use case "Retrieve Bicycle Lot Coordinates".
4. The application will retrieve the User's current coordinates from the GPS using the included use case "Get User Coordinates" every 10 seconds and display his/her current location on Google Maps.
5. If the User wants to find bicycle parking lots nearest to a specified destination, the application will prompt the User to input a destination.
6. The User will input a destination and use the filters "Shelter", and "Total Parking Lots".
7. The application will retrieve the destination coordinates from GoogleMaps using the included use case "Get Destination Coordinates" and display it on GoogleMaps.
8. The application compares the coordinates of the destination, with the coordinates of all bicycle parking lots, and returns the coordinates of the 5 closest bicycle parking lots by default, depending on the user's input in filters.
9. Suppose the User wants to find bicycle parking lots nearest to them based on their current location. In that case, the application will retrieve the User's current coordinates from the GPS using the included use case "Get User Coordinates" every 10 seconds.
10. The application compares the current coordinates of the User with the coordinates of all bicycle parking lots every 10 seconds the User's new coordinates are retrieved and returns the coordinates of the 5 closest bicycle parking lots.
11. The application displays the bicycle parking lots with GoogleMaps using the included use case "Display Bicycle Parking Lots".
12. The User can select a bicycle parking lot.

Includes Use case "Login", "Display Bicycle Parking Lots", "Retrieve Bicycle Parking Lot Coordinates", "Get User Coordinates", "Get Destination Coordinates"

Use Case 7: Report

Description: The user reports on a bicycle parking lot, whether it is non-existent or in poor condition.

Actors: User, Application System, GoogleMaps, Parking Lot Database, GPS

Pre-conditions: The User is logged into his/her account, with GPS enabled

Post-conditions: If the User selects a bicycle parking lot, the application will display all the reports on the selected bicycle lot by other Users

The flow of events:

1. The application will use the included use case “Login” to verify the User.
2. The User will use the included use case “Find Bicycle Parking Lot” to find a bicycle parking lot.
3. The User selects a bicycle parking lot and clicks on the option ‘Report’ to report the selected bicycle parking lot.
4. The application will prompt the User for a report response.
5. The User will input a string.
6. Once the User input is complete, the application will display the report on the selected bicycle parking lot by the User.
7. The User will also be able to see reports by other Users.

Includes Use case “Login”, “Find Bicycle Parking Lot”

Use Case 8: Add Destination

Description: User can save destinations that were previously searched for for a quick display of nearby bicycle parking lots

Actors: User, Application System, GoogleMaps

Pre-conditions: The User is logged into his/her account

Post-conditions: Destination coordinates are saved to the application, and the User will be able to find bicycle parking lots nearest to the saved destinations

The flow of events:

1. The application will use the included use case "Login" to verify the User.
2. To add a new destination to the list of saved destinations, the User can search for a location using the search bar on Google Maps.
3. The User selects the desired destination and clicks on the 'Save destination' option.
4. Google Maps will retrieve the destination coordinates and save them to the application system.
5. To access saved destinations, the User can click on the 'Saved destinations' option and select a saved destination.
6. The application will then find the nearest bicycle parking lots using the included use case "Find Bicycle Parking Lot".

Alternative flows:

AF-S3: If the selected destination is not within Singapore

1. The application will display "Invalid destination. Please try again!" for 1 second.
2. The application will return to step 2.

Includes Use case "Login", "Get Destination Coordinates", "Find Bicycle Parking Lot"

Use Case 9: Share Ride

Description: The user can share his ride destination, current location, and nearby bicycle parking lots with others via a hyperlink

Actors: User, Application System, Parking Lot Database, GPS, Google Maps

Pre-conditions: The User is logged into his/her account, with GPS enabled

Post-conditions: The location of the User, together with his ride destination will be shared with other users when they access the hyperlink

The flow of events:

1. The application will use the included use case "Login" to verify the User.
2. The User can find a bicycle parking lot using the included use case "Find Bicycle Parking Lot"
3. The location of the User, along with the ride destination will be updated and displayed on GoogleMaps using the included use case "Find Bicycle Parking Lot".
4. The locations of nearby bicycle parking lots will be updated and displayed if the User searches for the nearest bicycle parking lots based on his/her current location using the included use case "Find Bicycle Parking Lot".
5. If the search criteria of the User is based on a specified destination, the locations of the bicycle parking lots displayed on Google Maps will not change.
6. The user then clicks on the "Share Ride" option.
7. The application will generate a link to the map.
8. The User sends the link to other users.
9. Other users are now able to view the current location and the ride destination of the User by accessing the link.

Includes Use case "Login", "Find Bicycle Parking Lot".

Use Case 10: Group ride

Description: The host User can invite other users on a group ride via a hyperlink that shares the destination, nearby bicycle parking lots, and current locations of all users in the group

Actors: User, Application System, Parking Lot Database, GPS, Google Maps

Pre-conditions: The host User is logged into their accounts, with GPS enabled

Post-conditions: Ride destination, nearby bicycle parking lots and the current location of all users within the party are shared amongst each other

The flow of events:

1. The application will use the included use case “Login” to verify the host User.
2. The host User can find a bicycle parking lot using the included use case “Find Bicycle Parking Lot”
3. The location of the host User, along with the ride destination will be updated and displayed on GoogleMaps using the included use case “Find Bicycle Parking Lot”.
4. The locations of nearby bicycle parking lots will be updated and displayed if the host User searches for the nearest bicycle parking lots based on his/her current location using the included use case “Find Bicycle Parking Lot”.
5. If the search criteria of the host User is based on a specified destination, the locations of the bicycle parking lots displayed on Google Maps will not change.
6. The host User then clicks on the “Group Ride” option.
7. The application will generate a link to the map.
8. The host User sends the link to other users.
9. After other users access the link, the GPS will retrieve the current coordinates of all other users and display them on Google Maps.
10. All users within the group can view the current locations of everyone and the ride destination, together with bicycle parking lots nearest to the host User.

Includes Use case “Find Bicycle Parking Lot”, “Login”.

Use Case 11: Retrieve Bicycle Lot Coordinates

Description: Retrieve all bicycle parking lot coordinates from the Parking Lot Database

Actors: Parking Lot Database, User

Pre-conditions: There are coordinates of existing bicycle parking lots within the Parking Lot Database

Post-conditions: The application can retrieve all bicycle parking lot coordinates for the User to carry out the use case “Find Bicycle Parking Lot”

The flow of events:

1. When the User executes the use case “Find Bicycle Parking Lot”, the application will request for the data of all bicycle parking lots in the Parking Lot Database.
2. The Parking Lot Database will then return all data for each bicycle parking lot to the application, with the data consisting of lot coordinates, lot type, total capacity, and the presence of a shelter.
3. With the coordinates, the application can now compare them with the current coordinates of the User, or the destination coordinates, to complete the use case “Find Bicycle Parking Lot”.

Use Case 12: Get User Coordinates

Description: Retrieve current coordinates of the User using GPS

Actors: GPS, User

Pre-conditions: GPS is enabled

Post-conditions: The application can retrieve the current coordinates of the User to carry out the use case “Find Bicycle Parking Lot”

The flow of events:

1. When the User executes the use case “Find Bicycle Parking Lot”, the application will request the current coordinates of the User at every 10-second interval.
2. The GPS will get the current coordinates of the User and return them to the application at every 10-second interval.
3. With the User coordinates, the application can update and display the live location of the User, or find bicycle parking lots nearest to his/her current location, and complete the use case “Find Bicycle Parking Lot”.

Use Case 13: Get destination coordinates

Description: Retrieve the destination coordinates of the User using GoogleMaps

Actors: GoogleMaps, User

Pre-conditions: Destination input is valid

Post-conditions: The application can retrieve all bicycle parking lot coordinates for the User to carry out the use case “Find Bicycle Parking Lot”

The flow of events:

1. When the User executes the use case “Find Bicycle Parking Lot”, with the search criteria based on the destination input, the application will request the coordinates of the destination.
2. GoogleMaps will get the coordinates of the destination based on the User input and return them to the application.
3. With the destination coordinates, the application can now compare them with the coordinates of all bicycle parking lots, and complete the use case “Find Bicycle Parking Lot”.

Use Case 14: Display Bicycle Parking Lots

Description: Display nearby bicycle parking lots on GoogleMaps

Actors: GoogleMaps, User

Pre-conditions: Coordinates of nearby bicycle parking lots are retrieved from the Parking Lot Database

Post-conditions: The locations of nearby bicycle parking lots are displayed on GoogleMaps

The flow of events:

1. After the User executes the use case “Find Bicycle Parking Lot”, the application would have obtained the coordinates of the nearest bicycle parking lots.
2. The coordinates of the 5 bicycle parking lots will be passed to GoogleMaps, and displayed by default unless stated differently by the user.
3. The User will be able to view the location of the nearest bicycle parking lots on Google Maps.

Alternative flows:

AF-S3: If the User clicks on the “See more” option to view more bicycle parking lots

1. The application will display more locations of nearby bicycle parking lots on Google Maps using the extended use case “See More”.

Extends use case “See More Lots”, “See Lot Details”, “Directions”

Use Case 15: See More Lots

Description: The User can expand the view of the nearest bicycle parking lots from the nearest 5 to the next nearest 5 bicycle parking lots

Actors: GoogleMaps, User

Pre-conditions: Coordinates of nearby bicycle parking lots are retrieved from the Parking Lot Database

Post-conditions: The additional locations of nearby bicycle parking lots are displayed on GoogleMaps

The flow of events:

1. After the location of the nearest 5 bicycle parking lots is displayed on GoogleMaps after executing the use case “Find Bicycle Parking Lot”, the User clicks on the “See more” option.
2. The application will pass the coordinates of the next 5 nearest bicycle parking lots to GoogleMaps.
3. GoogleMaps will display the locations of the additional 5 bicycle parking lots.
4. The User will be able to view the location of the nearest 5 bicycle parking lots on Google Maps.

Alternative flows:

AF-S4: If the User decides to click on the “See more” option to view more bicycle parking lots

1. The application will return to Step 2 and then Step 5.
2. If all bicycle parking lots are displayed, clicking on the “See more” option will not display additional bicycle parking lots.

Use Case 16: See Lot Details

Description: The User can view all attributes of a bicycle parking lot, including its lot type, presence of shelter, and maximum capacity

Actors: GoogleMaps, User

Pre-conditions: Bicycle parking lots are displayed on GoogleMaps

Post-conditions: The User can view all attributes of bicycle parking lots displayed on GoogleMaps

The flow of events:

1. After the location of the nearest bicycle parking lots are displayed on GoogleMaps after executing the use case “Find Bicycle Parking Lot”, the User selects a bicycle parking lot.
2. The User clicks on the “See Lot Details” option.
3. The application will display all the attributes of the selected bicycle parking lot.

Use Case 16: Directions

Description: The User can view directions to the selected bicycle parking lot

Actors: GoogleMaps, User

Pre-conditions: Current coordinates of the User and bicycle parking lot coordinates are available on GoogleMaps

Post-conditions: The User can view directions to the selected bicycle parking lot from the current location on GoogleMaps

The flow of events:

1. After the locations of the nearest 5(default) bicycle parking lots are displayed on GoogleMaps after executing the use case “Find Bicycle Parking Lot”, the User selects a bicycle parking lot.
2. The User clicks on the “Directions” option.
3. Google Maps will show the directions from the current location of the User to the location of the selected bicycle parking lot.

Welcome!



Just checking if it's really you lol



SG Cycle Savvy



Pedal any weather.

Please verify your account

We just send a SMS to your mobile number +65 XXXX-XX99
with your verification code

Enter your one-time password

59s

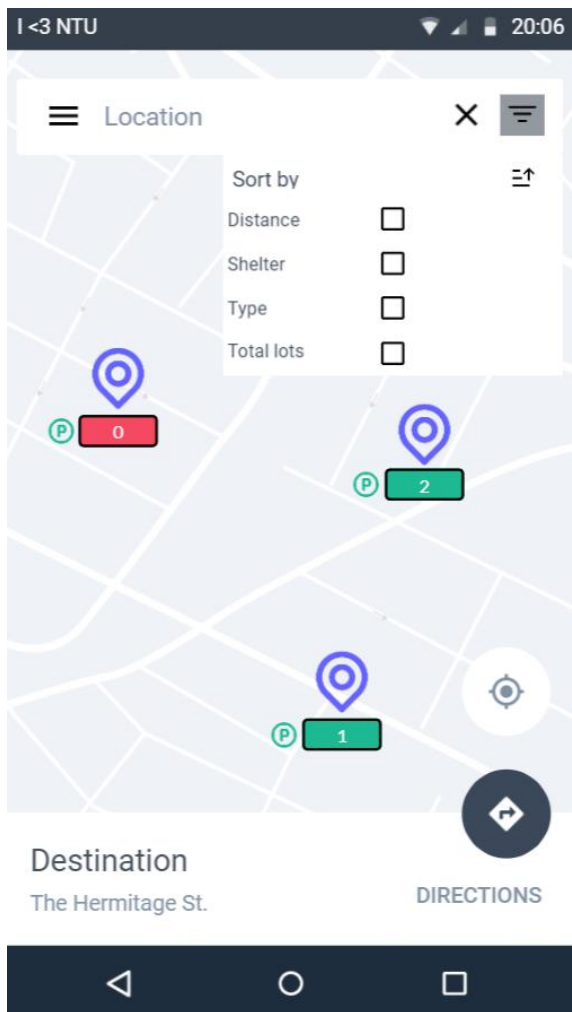
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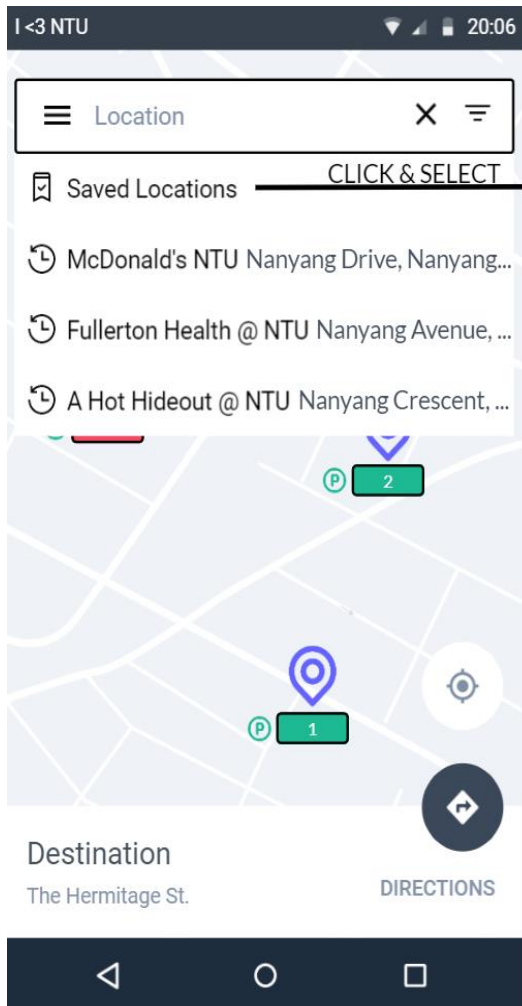
Password

LOG IN

LOG IN







Home Where I know I must be...

Work Pantry break, let's go!

In-laws' House Gives me nightmares...

Gym It's about drive, it's about?

